📐 EXAM REVISION GUIDE

SURVEYING

Unit II – Complete Formula Sheet

Basic Civil & Mechanical Engineering (BCME)

B.Tech 1st Year · 1st Semester

Definitions · Formulas · Instruments · Solved Problems · Flashcards

CHAPTERS	FLASHCARDS
10 Topics	**30 Q&A**

SOLVED SUMS	TABLES
3 Problems	**6 Tables**

📏 1. Objectives of Surveying

Primary Purposes

- Determine **relative positions** of points on earth's surface
- Prepare accurate **maps, plans & layouts** of land
- Set out engineering **alignments** (roads, canals, bridges)
- Calculate **areas & volumes** of land and earthwork
- Establish **control points** for construction projects
- Monitor **deformations & displacements** of structures

Role in Civil Engineering

- **Planning:** Site selection, feasibility study
- **Design:** Topographic data for drawings
- **Construction:** Setting out works on ground
- **As-built records:** Final surveys after completion
- **Legal:** Land boundary demarcation

> **Key Quote:** "Without surveying, no civil engineering project can be executed."

🌍 2. Basic Principles of Surveying

- **Whole to Part:** Start with large control network → divide into smaller areas (prevents error accumulation)
- **Two Measurements:** Every point fixed by minimum *two independent measurements*
- **Consistency:** All observations must be internally consistent
- **Check Every Measurement:** Never accept a single observation

Principle	Why Important
Whole to Part	Limits error propagation
Two Measurements	Cross-check accuracy
Accuracy first	Errors compound later
Independent checks	Detect mistakes early

📐 3. Horizontal Measurements

Definition: Measurement of distances projected on a horizontal plane between two points on the ground.

Methods of Linear Measurement

- **Direct:** Chain / Tape measurement
- **Optical:** Tacheometry, stadia method
- **Electronic:** EDM (Electronic Distance Measurement)
- **GPS:** Satellite-based positioning

Chain Surveying

- Simplest form of surveying

Instruments

Instrument	Use	Accuracy
Gunter's Chain	Distance measurement	Moderate
Steel Tape	Precise distance	High
Ranging Rod	Alignment / ranging	—
Arrows	Count chain lengths	—

- Only linear measurements — no angular
- Suitable for small, flat, open areas
- Network of triangles formed

Instrument	Use	Accuracy
Pegs	Mark station points	—
Offset Staff	Measure offsets	Low–Mod

Sources of Errors in Chaining

- **Personal:** Misreading, wrong counting of arrows
- **Instrumental:** Chain not standard length, kinks
- **Natural:** Temperature variation, slope, wind

Tape Survey – Advantages & Limits

- ✅ More accurate than chain
- ✅ Lightweight, easy to carry
- ❌ Breakable (invar tape)
- ❌ Affected by temperature (steel tape)
- ❌ Not suitable for rough terrain

4. Angular Measurements

Definition: Measurement of angles (horizontal or vertical) between survey lines using angular instruments.

Types of Angles

- **Horizontal Angle:** Angle between two lines in horizontal plane — used to determine direction and bearing
- **Vertical Angle:** Angle above (+ve) or below (−ve) horizontal — used in levelling & tacheometry
- **Deflection Angle:** Angle between extended previous line and next line
- **Included Angle:** Angle between two survey lines at a station

Instruments for Angular Measurement

Compass

Purpose: Measures magnetic bearings of lines

Principle: Freely suspended magnetised needle aligns with magnetic north

Types: Prismatic Compass, Surveyor's Compass

Low–Medium Accuracy

Theodolite

Purpose: Measures both horizontal & vertical angles precisely

Principle: Graduated circles + telescope + verniers

Applications: Traversing, triangulation, setting out

High Accuracy

5. Bearings — (HIGH IMPORTANCE)

Definition: The horizontal angle that a survey line makes with a reference direction (usually North), measured clockwise.

Types of Bearings

Type	Range	Format	Example
WCB (Whole Circle Bearing)	0° – 360°	One value	225°
QB (Quadrantal Bearing)	0° – 90°	N/S... °...E/W	S 45° W

WCB → QB Conversion

WCB Range	Quadrant	QB Formula
0° – 90°	NE (I)	QB = WCB
90° – 180°	SE (II)	QB = 180° – WCB

QB → WCB Conversion

QB	WCB Formula
N θ° E	WCB = θ
S θ° E	WCB = 180° – θ
S θ° W	WCB = 180° + θ
N θ° W	WCB = 360° – θ

Fore Bearing & Back Bearing

KEY FORMULA

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If FB < 180°: BB = FB + 180°
If FB > 180°: BB = FB − 180°
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WCB Range	Quadrant	QB Formula
180° – 270°	SW (III)	QB = WCB − 180°
270° – 360°	NW (IV)	QB = 360° − WCB

✏️ Solved Example 1 – WCB to QB Conversion

Convert the following WCB values to QB:

WCB	Range	Quadrant	Formula	QB Answer
55°	0–90	NE	QB = 55°	**N 55° E**
135°	90–180	SE	QB = 180–135 = 45°	**S 45° E**
225°	180–270	SW	QB = 225–180 = 45°	**S 45° W**
315°	270–360	NW	QB = 360–315 = 45°	**N 45° W**

✏️ Solved Example 2 – Fore Bearing & Back Bearing

Given: FB of line AB = 70°. Find BB of AB and QB equivalents.

- FB (WCB) = 70° (in NE quadrant → QB = N 70° E)
- Since FB = 70° < 180°: BB = 70° + 180° = **250°**
- BB = 250° (in SW quadrant → QB = 250° − 180° = 70° → **S 70° W**)

📊 6. Levelling – Definitions & Terms

Levelling: The art of determining the relative heights (elevations) of different points on the earth's surface, or of fixing points at given heights above a datum.

Purpose: To find difference in elevation between points for drainage design, road grades, dam sites, building foundations.

Term	Symbol	Definition	Notes
Datum	—	Reference surface (MSL in India)	RL of datum = 0
Benchmark (BM)	BM	Fixed point with known RL	GTS BM, permanent BM, temporary BM
Reduced Level	RL	Height above datum	RL = HI − Staff reading
Height of Instrument	HI	RL of line of collimation	HI = RL of BM + BS
Back Sight	BS	First reading after setting level	Always on known RL point
Fore Sight	FS	Last reading before shifting level	Always on change point
Intermediate Sight	IS	Reading on intermediate points	Between BS and FS
Change Point	CP	Point where both BS & FS taken	Staff not moved during reading
Line of Collimation	—	Line through optical axis of telescope	Must be truly horizontal
Rise	R	Previous RL < Current RL	+ve difference
Fall	F	Previous RL > Current RL	−ve difference

🔧 7. Levelling Instruments

🔭 Dumpy Level

Function: Provides horizontal line of sight

Principle: Telescope + bubble tube, both fixed to vertical axis

Parts: Telescope, bubble tube, tribrach, foot screws, diaphragm

Setting up: Temporary adjustment → centering → levelling → focusing

Primary Instrument

📏 Levelling Staff

Function: Graduated rod held vertically at points whose RL is required

Types: Solid staff, Folding staff, Telescopic staff

Reading: E-type (4.0m range), Sopwith (5m)

Accuracy: Read to nearest 0.005 m

Secondary Instrument

📐 Tripod

Function: Provides stable base for the level

Types: Fixed head, Sliding-leg, Telescopic

Material: Wood or aluminum

Setup: Legs spread equally, firmly planted on ground

Support Equipment

8. Levelling Methods & Formulas

Method 1: Height of Instrument (HI) Method

HI METHOD FORMULAS

HI = RL of BM + BS

RL of point = HI − Staff Reading

RL (IS) = HI − IS
RL (FS) = HI − FS

Arithmetic Check (HI Method)

VERIFICATION FORMULA

ΣBS − ΣFS = Last RL − First RL

- ✅ Simple and fast
- ✅ Best when many IS readings at same HI
- ❌ No check on IS readings

Method 2: Rise & Fall Method

RISE & FALL FORMULAS

Rise = Previous reading − Current reading (if +ve)

Fall = Current reading − Previous reading (if +ve)

RL = Previous RL + Rise (or − Fall)

Arithmetic Check (R&F Method)

VERIFICATION FORMULA

ΣBS − ΣFS = ΣRise − ΣFall = Last RL − First RL

- ✅ Complete check on every reading
- ✅ More accurate, error easily detected
- ❌ More time-consuming calculations

Feature	HI Method	Rise & Fall Method
Speed	Faster	Slower
Arithmetic check	Partial (no IS check)	Complete (all readings checked)
Best for	Many IS at same setup	Precise levelling
Error detection	Poor for IS	Excellent
Preferred in	Longitudinal sections	Precise/geodetic levelling

✏️ Solved Levelling Problem

Problem: The following staff readings were observed (BM RL = 100.000 m). Solve by BOTH methods.

Readings: 2.500 (BS on BM), 1.800 (IS), 1.200 (IS), 0.900 (FS/CP), 2.200 (BS), 1.500 (IS), 0.600 (FS)

📋 HI Method Solution:

Station	BS	IS	FS	HI	RL	Remarks
BM	2.500	—	—	102.500	100.000	BM (given)
A	—	1.800	—	—	100.700	IS
B	—	1.200	—	—	101.300	IS
CP	2.200	—	0.900	103.800	101.600	Change Point
C	—	1.500	—	—	102.300	IS
D	—	—	0.600	—	103.200	Last Point
Σ	**4.700**	—	**1.500**	—	—	—

> **Check:** ΣBS − ΣFS = 4.700 − 1.500 = 3.200 | Last RL − First RL = 103.200 − 100.000 = 3.200 ✅

📋 Rise & Fall Method Solution:

Station	BS	IS	FS	Rise	Fall	RL
BM	2.500	—	—	—	—	100.000
A	—	1.800	—	0.700	—	100.700
B	—	1.200	—	0.600	—	101.300
CP	2.200	—	0.900	0.300	—	101.600
C	—	1.500	—	0.700	—	102.300
D	—	—	0.600	0.900	—	103.200
Σ	**4.700**	—	**1.500**	**3.200**	**0**	—

> **Check:** ΣBS − ΣFS = 3.200 | ΣRise − ΣFall = 3.200 | Last RL − First RL = 3.200 ✅ All three match!

🗺️ 9. Contour Mapping

Contour: An imaginary line on the ground connecting all points having the same reduced level (elevation).

Contour Interval (CI): Vertical distance between two consecutive contour lines. Smaller CI = more detail, but more lines.

Characteristics of Contour Lines

- Contours are always **closed curves** (may close outside map boundary)
- Two contours of **different elevations never cross** (except in caves / overhangs)
- **Closely spaced** = steep slope; **widely spaced** = gentle slope
- Contours are **perpendicular to direction of steepest slope**
- Contours **crossing a ridge** form a V-shape pointing downhill
- Contours **crossing a valley** form a V-shape pointing uphill
- A **hill** is shown by closed contours, inner = higher
- A **depression** = closed contours, inner = lower (hachured)

Uses of Contour Maps

- **Site selection** for buildings, dams, reservoirs
- **Route alignment** for roads, railways, canals
- **Water flow direction** and catchment area
- **Intervisibility** between two stations
- **Cross-section** drawing and earthwork calculation
- **Slope analysis** for construction planning

Types of Contours

Feature	Contour Shape
Hill / Knoll	Concentric closed curves (inner higher)
Valley	V-shapes pointing uphill
Ridge	V-shapes pointing downhill
Cliff	Nearly touching contours
Depression	Closed with hachures
Plain	Widely spaced

10. Important Comparison Tables

Table 1: Surveying Methods Comparison

Method	Measurement Type	Key Instruments	Accuracy	Best Used For
Chain Surveying	Linear only	Chain, Ranging rods	Low–Moderate	Small, flat, open areas
Compass Surveying	Linear + Angular	Prismatic compass, Chain	Moderate	Forested or rough terrain
Plane Table Surveying	Angular (graphical)	Plane table, Alidade	Moderate	Small-scale mapping
Theodolite Surveying	Angular (precise)	Theodolite, Tape	High	Traversing, triangulation
Levelling	Vertical (elevation)	Dumpy level, Staff	High	Finding RL of points
Tacheometry	Linear (optical)	Tacheometer	Moderate	Hilly/inaccessible terrain
EDM / Total Station	Electronic	Total Station	Very High	Large scale precise work

Table 2: Instruments Comparison

Instrument	Primary Purpose	Used In	Accuracy Level
Gunter's Chain (20m)	Measuring distances	Chain surveying	Low
Steel Tape	Precise distance measurement	All types	High
Invar Tape	Very precise baselines	Triangulation	Very High
Ranging Rod	Alignment of chain line	Chain/tape survey	—
Prismatic Compass	Magnetic bearings	Compass traversing	Low–Moderate
Theodolite	Horizontal & vertical angles	Traversing, setting out	High
Dumpy Level	Horizontal line of sight	Levelling	High
Levelling Staff	Staff readings at points	Levelling	High
Plane Table	Graphical field plotting	Plane table survey	Moderate
Total Station	Angles + distances (electronic)	All engineering surveys	Very High

Table 3: Bearings Comparison

Type	Full Form	Range	Notation	Example
WCB	Whole Circle Bearing	0° to 360°	Single angle	135°
QB	Quadrantal Bearing	0° to 90° each quadrant	N/S θ° E/W	S 45° E
True Bearing	Measured from True North	0° to 360°	Degrees	270° T

Type	Full Form	Range	Notation	Example
Magnetic Bearing	Measured from Magnetic North	0° to 360°	Degrees	268° M
FB	Fore Bearing	Any	A to B direction	60°
BB	Back Bearing	FB ± 180°	B to A direction	240°

Table 4: Sources of Error in Surveying

Type	Definition	Examples	How to Minimize
Personal Errors	Due to human limitations	Misreading staff, poor estimation	Practice, multiple readings
Instrumental Errors	Due to imperfect instruments	Chain not standard, bubble not centered	Regular calibration, adjustment
Natural Errors	Due to environment	Temperature, wind, refraction	Corrections, favorable weather
Systematic Errors	Follow definite law	Tape sag, temperature correction	Apply corrections
Random Errors	No fixed pattern	Eye fatigue, vibration	Statistical methods, repetition

🧠 11. All Formulas & Key Relations

📐 CHAIN/TAPE SURVEYING

Horizontal Distance = $L \cos \theta$
(L = slope distance, θ = slope angle)

Correction for slope = $L(1 - \cos \theta)$

Actual length = Measured length × (True length / Nominal length)

🧭 BEARINGS

WCB + 180° = Back Bearing (if WCB < 180°)
WCB − 180° = Back Bearing (if WCB > 180°)

Declination = True Bearing − Magnetic Bearing

If declination East (+ve): True = Magnetic + δ
If declination West (−ve): True = Magnetic − δ

📐 TRAVERSE CALCULATIONS

Departure = $L \times \sin \theta$
Latitude = $L \times \cos \theta$

Easting (+ve departure), Westing (−ve)
Northing (+ve latitude), Southing (−ve)

Closing Error = $\sqrt{(\Sigma D^2 + \Sigma L^2)}$
Precision = Closing error / Perimeter

📊 LEVELLING – HI METHOD

HI = RL(BM) + BS

RL(point) = HI − Staff reading

Arithmetic Check:
$\Sigma BS - \Sigma FS$ = Last RL − First RL

📉 LEVELLING – RISE & FALL METHOD

Rise = Previous staff reading − Current staff reading (if +ve)

Fall = Current staff reading − Previous staff reading (if +ve)

RL = Previous RL + Rise (or − Fall)

Arithmetic Check:
$\Sigma BS - \Sigma FS = \Sigma Rise - \Sigma Fall$ = Last RL − First RL

🗺 CONTOUR – GRADIENT FORMULA

Gradient = Contour Interval / Horizontal Distance

Horizontal Equivalent (HE) = CI / $\tan \theta$
(θ = angle of slope)

Scale: Map distance = Ground distance / Scale factor

⚡ QUICK FORMULA MEMORY AID

HI = BM + BS | RL = HI − Staff | BB = FB ± 180° | $\Sigma BS - \Sigma FS$ = Last RL − First RL

🧠 12. Flashcards – Quick Q&A (30 Cards)

Q: What is surveying?
A: Measurement & mapping of relative positions of points on earth's surface

Q: What is a bearing?
A: Horizontal angle a line makes with reference direction (North), measured clockwise

Q: What is RL?
A: Reduced Level — height of a point above the datum surface (MSL)

Q: What is a Benchmark?
A: A fixed point of known RL used as reference in levelling surveys

Q: What is WCB?
A: Whole Circle Bearing — angle 0° to 360° measured clockwise from North

Q: What is QB?
A: Quadrantal Bearing — angle 0–90° with N or S as reference, towards E or W

Q: Define Back Sight (BS)
A: First staff reading after setting up the level — taken on a known RL point

Q: Define Fore Sight (FS)
A: Last reading before shifting the level — taken on a change point

Q: What is Height of Instrument (HI)?
A: RL of the line of collimation: HI = RL(BM) + BS

Q: What is a Change Point?
A: Point where both BS and FS readings are taken — staff not moved

Q: WCB of S 45° W in QB?
A: 225° (WCB = 180° + 45°)

Q: FB of AB = 250°. Find BB.
A: BB = 250° – 180° = 70°

Q: What is a contour line?
A: Imaginary line joining all points of equal elevation on ground

Q: Closely spaced contours indicate?
A: Steep slope

Q: What is contour interval?
A: Vertical difference between two consecutive contour lines

Q: Two contours cross — what does that mean?
A: Cave or overhanging cliff (rare exception)

Q: What is a Dumpy Level?
A: Levelling instrument providing horizontal line of sight using bubble tube + telescope

Q: Purpose of Ranging Rods?
A: To align chain/tape along a straight survey line

Q: What is the principle of surveying?
A: Work from whole to part; every point fixed by at least two measurements

Q: What is a traverse?
A: A series of connected survey lines whose lengths and directions are measured

Q: What is intermediate sight (IS)?
A: Staff reading taken between BS and FS at any intermediate point

Q: What is the datum in India?
A: Mean Sea Level (MSL) at Karachi (historically); Mumbai nowadays used

Q: What is magnetic declination?

A: Horizontal angle between true north and magnetic north

Q: Rise in levelling = ?

A: When previous staff reading is greater than current reading (elevation increases)

Q: Check for Rise & Fall method?

A: $\Sigma BS - \Sigma FS = \Sigma Rise - \Sigma Fall = \text{Last RL} - \text{First RL}$

Q: What is tacheometry?

A: Optical method of measuring distance using stadia lines on theodolite

Q: Chain surveying uses what type of measurements?

A: Linear measurements only (no angular measurements)

Q: Theodolite measures?

A: Both horizontal and vertical angles with high precision

Q: What does GTS BM stand for?

A: Great Trigonometrical Survey Benchmark — most permanent type of BM in India

Q: V-shape contours pointing uphill means?

A: Valley or stream channel

⚡ 13. Exam Tips & Common Mistakes

❌ Common Mistakes to Avoid

❌ MISTAKE 1 – CONFUSING WCB AND QB

In WCB, always measure CLOCKWISE from North only. In QB, you need to prefix N/S and suffix E/W. Never write QB as just "45°" — always write "N 45° E" etc.

❌ MISTAKE 2 – WRONG BACK BEARING FORMULA

If FB < 180° → add 180°. If FB > 180° → subtract 180°. Many students always add 180° regardless of the FB value.

❌ MISTAKE 3 – HI METHOD ARITHMETIC CHECK

Check is only ΣBS − ΣFS = Last RL − First RL. IS readings are NOT included in this check!

❌ MISTAKE 4 – RISE/FALL CALCULATION DIRECTION

Rise = Previous reading MINUS current reading. Students often reverse this. Remember: smaller staff reading = higher ground.

❌ MISTAKE 5 – CONTOUR LINE CROSSING

Two different-elevation contours NEVER cross on normal ground. If you draw them crossing, it's wrong (except caves).

✅ Exam Writing Tips

💡 TIP 1 – ALWAYS SHOW ARITHMETIC CHECK

In any levelling problem, ALWAYS write the arithmetic check at the end. It earns full marks and shows the examiner you know the method completely.

💡 TIP 2 – DRAW TABLES FOR LEVELLING

Never write levelling calculations in running prose. Always use the standard BS/IS/FS table format — examiners look for this.

💡 TIP 3 – STEP-BY-STEP BEARING CONVERSION

Write: Step 1 – Identify quadrant, Step 2 – Apply formula, Step 3 – Write final QB/WCB. This earns step marks even if final answer has error.

💡 TIP 4 – DRAW ASCII DIAGRAM FOR CONTOURS

For contour questions, sketch a simple diagram showing hill/valley with labeled contours. A small sketch can fetch 2–3 extra marks.

💡 TIP 5 – DEFINE BEFORE SOLVING

For any problem-based question, write 1-line definition of the method before solving. Example: "HI Method: RL = HI − Staff reading, where HI = RL(BM) + BS"

🎯 **High-Value Topics for Exam:** Bearing conversions (WCB ↔ QB, FB ↔ BB) | Levelling numericals (HI + R&F both methods) | Contour characteristics | Instrument names and purposes | Definitions of levelling terms

🚀 14. Quick Revision Sheet – Last 30 Minutes

📌 KEY DEFINITIONS

- Surveying = Map positions on earth
- Bearing = Angle from North (clockwise)
- RL = Height above datum

🔢 KEY FORMULAS

- HI = RL(BM) + BS
- RL = HI − Staff reading
- BB = FB ± 180°
- WCB→QB: Subtract 180° / 360°

🔧 INSTRUMENTS

- Chain → Linear distance
- Steel Tape → Precise distance
- Ranging Rod → Alignment
- Prismatic Compass → Bearings

- BM = Fixed point, known RL
- HI = RL of line of sight
- Contour = Equal elevation line
- CI = Vertical gap between contours
- BS = First reading (known pt)
- FS = Last reading (change pt)
- IS = Middle readings

- QB→WCB: Add 180° / 360°
- Rise: Prev − Current (+ve)
- Fall: Current − Prev (+ve)
- Check: ΣBS−ΣFS = ΔRL
- Dep = L sin θ
- Lat = L cos θ

- Theodolite → H+V angles
- Dumpy Level → Levelling
- Levelling Staff → Staff readings
- Plane Table → Field plotting
- Total Station → All-in-one
- Tripod → Support base

🧭 BEARING QUICK TABLE

- 0–90° → NE quad (QB = N θ° E)
- 90–180° → SE quad (QB = S (180−θ)° E)
- 180–270° → SW (QB = S (θ−180)° W)
- 270–360° → NW (QB = N (360−θ)° W)
- FB < 180° → BB = FB + 180°
- FB > 180° → BB = FB − 180°

📊 CONTOUR RULES

- Always closed curves
- Never cross (except caves)
- Close spacing = steep slope
- Wide spacing = gentle slope
- V → uphill = valley
- V → downhill = ridge
- Hill: inner contour higher
- Depression: hachures shown

✅ ARITHMETIC CHECKS

- HI Method: ΣBS−ΣFS = ΔRL
- R&F Method: ΣBS−ΣFS = ΣR−ΣF = ΔRL
- All three values must match
- BS always on known/BM point
- FS always on change point
- IS readings: no direct check (HI)
- IS readings: fully checked (R&F)

🏆 **Topper Formula:** In exam, structure every answer as: *Definition → Formula/Instrument → Example/Table → Sketch → Check*. This systematic approach guarantees maximum marks!